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Reflective Journal

In this lab, I learned how exploratory data analysis (EDA) helps us better understand a dataset before performing more advanced analysis. The Titanic dataset provided a good example because it contains different types of information about passengers, including age, gender, ticket fare, passenger class, and survival status. By exploring the data, I was able to identify patterns and relationships that would not be obvious by simply looking at rows of numbers.

One of the most useful parts of the lab was creating visualizations. The charts and graphs made it easier to see trends in the data. For example, I observed that female passengers had a higher survival rate than male passengers. I also noticed that passengers in higher classes generally had better survival outcomes. These visualizations helped me understand how data can be used to answer questions and support conclusions.

The experiments in the lab also helped me think critically about relationships between variables. Looking at factors such as port of embarkation and ticket fare allowed me to explore additional patterns in passenger survival. While some relationships were stronger than others, the process showed me how data analysis can reveal meaningful insights.

Overall, this lab improved my understanding of exploratory data analysis and the importance of visualizing data before making decisions. I gained more confidence using Python, pandas, and data visualization tools to analyze datasets. These skills will be valuable in future data science and artificial intelligence projects because they help transform raw data into useful information that can support decision-making.

Ved Dalwadi

In this journal entry, I will share my thoughts about the Module 4 exercise and Exploratory Data Analysis. After learning how to train machine learning models, it is important to know that the data used to train the model is just as important as, or more important than, the model itself. Using the Titanic dataset, I learned how to use visualizations to find patterns shown that were common at that time. The purpose of this reflection is to explain EDA and how it helps in understanding the dataset.

The lab showed that EDA is the detective work of the machine learning process. Before this, I believed that the mean or median was enough information, but after EDA, I realized that this was scratching the surface. Using the `describe` command gave me averages like passenger age or overall survival rate of 38%; however, after I visualized the data, I realized that these numbers won't show the full story. Using the `countplot` showed me that survival wasn't just a number based on luck but was dependent on other factors. For example, the class plot showed that first-class passengers had survived much better. The gender data showed the same pattern for women, as they survived at a much higher rate. Seeing this, it made it clear why visualizing matters, as 38% may have survived, but it shows that class and gender make a big difference.

The part of the lab I felt was most difficult was the experimentation section, where I was told to code on my own. For the boarding by port experiment, I noticed that Southampton has the most passengers but the most deaths as well. While Cherbourg was the only port where there were more survivors than deaths. I didn't believe the port had anything to do with the survival rate; however, I did believe the people did. After further research, I understood that Cherbourg had a very wealthy population, as it was where the global elite connected. This showed me that EDA isn't about spotting a pattern but wondering if that's the real reason behind the data. The fare boxplot backed up my research, as only the wealthy were most of the survivors because of status over the commoners and proximity to the lifeboats.

As I worked on the histograms comparing the age of survivors to non survivors matched the women and children first idea that was mentioned in the notebook. It wasn't as direct as class or gender, but it still shows the importance of EDA in machine learning.

This lab helped me understand EDA and once again showed me that data preparation is the most important part of machine learning. If I had made a model based on the raw data without looking at it I would have caused many issues and would have been much more of a hassle to deal with while also training an AI. Moving on I want to better understand the data I'm given and want to understand how the AI will perceive it.

Nigel Bontiff

Reflective Journal: Lab 04 Exploratory Data Analysis

This lab helped me understand Exploratory Data Analysis in a more practical way. Before

this assignment, I understood EDA mostly as “looking at the data before modeling,” but that felt kind of vague. After working with the Titanic dataset, I can see that EDA is really about slowing down and getting familiar with the dataset before trying to make predictions from it. It is almost like checking the condition of the data before trusting it. One of the first things that stood out to me was how much information came from simple commands like `df.head()`, `df.info()`, and `df.describe()`. At first, these seemed like basic setup commands, but they actually gave me a useful overview of the dataset. I could see the columns, the data types, the number of passengers, the average age, the survival rate, and which columns had missing values. The missing values were especially important because Age, Cabin, and Embarked were not completely filled in.

The survival distribution graph helped me understand class imbalance better. Seeing that more passengers died than survived made the idea more concrete. I had heard that imbalanced data can be a problem for machine learning, but the chart made it easier to see why. If a model is trained on data where one outcome happens much more often than another, it might lean too heavily toward predicting the majority class. That is something I probably would not have thought about as much if I had only looked at the raw table.

The passenger class and gender charts were probably the clearest parts of the lab for me. Passenger class showed a strong pattern, where first class passengers had a much better survival outcome than third class passengers. The gender chart was even more obvious, with females surviving at a much higher rate than males. These graphs helped me understand why visual EDA matters. The patterns were technically in the data the whole time, but the plots made them much easier to notice and explain.

The age distribution section also helped me think about how different plots answer different questions. The side by side histograms made it easier to compare passengers who survived with passengers who did not. I noticed that many passengers who did not survive were adults, while the survival group had a visible number of young children. I would not say age explains everything, but the graph does suggest that age may have played some role. This reminded me that EDA is not about proving a final answer right away. It is more about finding patterns and asking better questions.

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The student experiments helped me practice that same idea. For embarkation port, passengers from Cherbourg had the highest survival rate. At first, it would be easy to say that the port caused better survival, but I do not think that is the right conclusion. It might be connected to other factors, like passenger class or fare. That was a useful lesson for me because EDA can show relationships, but it does not automatically prove cause and effect.

The fare analysis also stood out because the boxplot and summary statistics pointed in the same direction. Passengers who survived generally paid higher fares than passengers who did not survive. That connects back to passenger class, because higher fares probably overlap with first class tickets. This made me realize that features in a dataset are not always totally separate from each other. Sometimes different columns are telling parts of the same story.

One challenge I had was dealing with the code formatting. Some of the code got pasted with multiple commands on one line, which caused syntax errors. Fixing that helped me understand Python a little better. I learned that even if the logic is correct, Python still needs the structure to be clean. Imports, print statements, and plotting commands usually need their own lines. It was a small issue, but it made me pay more attention to how Python reads code.

Overall, this lab changed how I think about EDA. I used to see graphs and summary statistics as something extra, but now I see them as part of the actual analysis. Before building a model, I need to understand the data, check for missing values, look for patterns, and think about whether the relationships make sense. The biggest thing I took away is that EDA keeps you from modeling blindly. It gives you a better idea of what the data is saying and what problems might come up later.